

Aryan Miriyala

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PROFESSIONAL SUMMARY

Computer science graduate and software engineer with experience across full-stack product engineering, applied AI systems, cloud data engineering, secure workflow modernization, cybersecurity education, manufacturing/process-improvement technology, and systems research. Built AI/model gateway infrastructure, customer/admin portals, LLM/RAG workflows, AI-assisted products, AWS data pipelines, database automation, secure internal tools, healthcare workflow improvements, and C++/OpenMP runtime experiments using Python, TypeScript/JavaScript, SQL, React, Next.js, Firebase, Angular, Node.js, PostgreSQL, MongoDB, MySQL, Docker, AWS, PySpark, LiteLLM, LangChain, OpenAI API, Gemini API, Mistral AI, and ChromaDB.

EDUCATION

Bowling Green State University

Master of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH

Aug. 2024 – Aug. 2026

Bowling Green State University

Bachelor of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH

Jan. 2021 – Apr. 2024

EXPERIENCE

Software Engineering Intern

Actual Reality Technologies

Jul. 2026 – Present

Perrysburg, OH

- Deployed AI Gateway routing and safety infrastructure around a pinned LiteLLM proxy on Railway, enabling intelligent model routing across multiple AI projects with smart-router tiers, guardrails, Langfuse traces, Slack alerts, config validators, CI/deploy gates, and red-team fixtures.
- Building a remote MCP server with keyed access for the Next.js/Firebase customer portal so customer/admin AI tools can query Plane ticket/project status and submit intake requests with scoped security and cost controls.
- Consolidated customer workflow tools into a feature-gated tabbed portal experience with legacy redirects, reusable tab UI, accessibility/navigation-history guardrails, route tests, component tests, and documentation updates.
- Hardened customer-portal project-data reliability by paginating work-item state counts, forcing cache-bypassing refreshes through route/gateway layers, preventing stale overlapping reads, and deriving project states from issue counts instead of rounded progress.
- Improved production authentication and session behavior by deriving cookie domains from request hosts, stabilizing password and secure account-link login transitions, and working through follow-up account-lifecycle risks around account creation, role-change side effects, invite domains, and Auth/user-document drift.
- Added and extended Vitest, Testing Library, route, component, architecture, and health-check coverage for project-management API integration, dashboard semantics, auth/session behavior, membership remediation, revoked write access, Firestore read-error visibility, and customer workflow validation.

Artificial Intelligence & Computer Science Intern

SmartSolve Industries

May 2026 – Jul. 2026

Bowling Green, OH

- Deployed role-specific AI agents for employee-requested workflows and converted tool audits into scoped projects, helping operations and executive leaders prioritize internal AI investments.
- Architected a full-stack onboarding tracker with Next.js, TypeScript, PostgreSQL, Drizzle ORM, SSO, and auth middleware, replacing memory-based HR follow-up with secure new-hire handoff tracking.
- Modeled an AI-enabled QMS dependency register to make quality-document relationships traceable before process changes, reviews, and downstream implementation work.
- Built a cross-platform Docker devcontainer for WSL2 and Colima, improving reproducible setup and isolating AI-assisted development workflows around proprietary source code.
- Evaluated LLM/API and cloud-service integration paths for internal workflow automation, including lightweight proprietary tools, API integrations, webhook-based automations, internal knowledge-base documentation, and applied business-problem training material.
- Applied Codex and Claude Code for planning, implementation, debugging, code review, and self-review while keeping human validation central around proprietary source code and company data.

Graduate Research Assistant I

Bowling Green State University

Aug. 2024 – May 2026

Bowling Green, OH

- Developed cybersecurity education materials, lab guides, workshops, and presentations across ethical hacking, OSINT, web security, email security, cryptography, secure key management, and cyber workforce topics.

- Built hands-on instructional content using Kali Linux, Nmap/NSE, Metasploit, OpenVAS/Greenbone, Wireshark/tcpdump, Burp Suite, OWASP ZAP, Shodan, Netcat/Ncat, Snort, John the Ripper, Hydra, DVWA, Metasploitable2, Vulnserver, and PentesterLab.
- Mapped NICE Framework and CAE-CD curriculum requirements into student- and instructor-facing materials covering cybersecurity roles, knowledge areas, tools, and instructional requirements.
- Created email security and forensics material covering phishing analysis, email headers, SPF/DKIM/DMARC, PGP/GnuPG, packet captures, and secure email communication.
- Built cryptography and secure key-management content involving symmetric/asymmetric encryption, hashing, PKI, key storage, key recovery, Azure Key Vault lab material, and Shamir's Secret Sharing.
- Led 10 undergraduate students in organizing a middle-school CyberCamp using CyberGuardian modules, cryptography activities, phishing demos, password-security exercises, and cyber escape-room material.

Data Engineering Intern

May 2025 – Aug. 2025

American Association of Insurance Services

Lisle, IL

- Automated a weekly DBEaver SQL billing process with Python, PySpark, and AWS Glue, processing 20+ TB of production insurance data for Data Solutions, internal teams, and CFO reporting.
- Architected a 160+ table MDM migration by profiling MySQL, Oracle, and Impala sources with Python, Pandas, SQL, and JDBC, standardizing inconsistent legacy insurance data into 25 domains.
- Modernized 160+ legacy Pentaho jobs into AWS Glue ETL workflows and partitioned S3 pipelines for Semarchy MDM, improving maintainability for governed MDM ingestion.
- Delivered reverse-ETL and data-validation workflows with custom AWS IAM roles, giving users controlled self-service access to validated insurance data while maintaining 24-hour latency.
- Connected billing, MDM, reverse-ETL, validation, and self-service workflows across AWS Glue, S3, IAM, SQL, JDBC, and Semarchy MDM/xDM, improving governed insurance-data availability.

Software Engineering Intern

May 2023 – May 2024

American Association of Insurance Services

Lisle, IL

- Engineered Python/JSON automation that generated 1,000+ production SQL tables and consolidated views across 10+ insurance lines, accelerating openIDL/openIDS-aligned data standardization and reporting.
- Automated CSV cleaning and deduplication with Python, Pandas, multiprocessing, fuzzy matching, and hashing, improving source quality for insurance standardization pipelines.
- Implemented RBAC and JWT patterns in a React/Node.js modernization, replacing broad employee-wide visibility with role-specific controls for about 110 internal users.
- Automated AWS Lambda validation that tokenized PII with hashlib/boto3 and delivered structured JSON to S3, supporting safer downstream insurance data processing for SDMA workflows.
- Combined Python/JSON schema generation, React/Node.js access controls, and AWS Lambda/S3 validation to modernize insurance data-standardization workflows.

Software Engineering Intern

May 2022 – Aug. 2022

Alliance for Paired Kidney Donation

Toledo, OH

- Secured AWS-hosted Lucee/CFML match-run workflows with anti-CSRF, IP-aware audit logging, protected routing, crash fixes, and blood-tube defaults for safer transplant operations.
- Implemented global anti-CSRF form protection, IP-aware audit logging, protected access-denied routing, and clearer 404 behavior across internal workflows.
- Modernized legacy ColdFusion/Lucee forms with HTML5 date/time pickers, themed CSS, JavaScript fallbacks, cform script updates, and graceful degradation for manual input.
- Improved transplant logistics and data reliability by adding center-level blood-tube defaults, logistics-sheet population, Matchgrid upload ordering, MFI file-location fixes, and duplicate-antigen MFI crash handling.
- Refactored MatchRun, registration, offer, and combination interfaces into reusable modal/tab components with icons and tooltips to improve usability and maintainability.

PROJECTS

Travel Health Advisor - BGSU Hackathon 2025

2025

- Built a full-stack health AI prototype with React, Vite, Express, MongoDB/Mongoose, and Mistral AI for travel-health guidance.
- Implemented an interactive country-selection map with react-simple-maps, retrieving disease records from an Express/MongoDB API and surfacing risk, vaccine availability, healthcare access, recovery-rate data, and travel recommendations.
- Built a health-profile form and context-aware Mistral AI chatbot that used saved user health details to answer travel-health questions.

- Generated downloadable vaccination checklist PDFs with @react-pdf/renderer based on country-specific vaccine-required disease data.

Fix-It-Flow - RocketHacks 2026

2026

- Built a Next.js 14/TypeScript PWA that uses camera input, voice commands, Gemini Vision, Featherless.AI/Llama reasoning, ElevenLabs TTS, and DynamoDB to guide appliance repair and recycling decisions.
- Implemented a voice-first inspection workflow combining live-frame analysis, spoken user context, conversation history, structured findings, confidence scores, requested camera views, and safety warnings.
- Designed hands-free repair guidance with AI-generated steps, ElevenLabs audio playback, contextual Q&A, and voice commands for next, repeat, help, and end session.
- Built typed Next.js API routes and AWS-backed session persistence for inspection turns, repair sessions, device identification, text-to-speech, chat, uploads, findings, frames, and repair steps.

Diff-Grounded Pull Request Description Generation with Structured Evidence

2026

- Developed an ICSME-published LLM pipeline that turns commits, diffs, linked issues, and repository metadata into evidence-grounded PR descriptions, reducing reviewer effort across 1,450 PRs.
- Developed structured PR-context construction, weak-commit-message improvement, and file-level summarization using Python, OpenAI API, and GitHub API to keep generated descriptions traceable to source artifacts.
- Evaluated generated PR descriptions with LLM-as-judge and human review, showing stronger grounding and reviewer preference than current AI-generated and human-written baselines.

DreamScape: AI-Powered Sleep Learning Companion

- Built a cross-platform React Native/Expo mobile app for sleep-based microlearning with AI-generated study cues, quizzes, text-to-speech audio, and local-first session tracking.
- Integrated Gemini and ElevenLabs APIs to summarize PDF/TXT documents, generate quizzes, synthesize learning cues, cache audio, and schedule simulated sleep-mode playback.
- Designed typed Zustand/AsyncStorage stores for topics, flashcards, cues, sleep sessions, quiz results, and user settings in a local-first mobile architecture.

FalconGraph Search: AI-Powered Campus Knowledge Search

- Built a BGSU campus answer engine that integrated a multithreaded C++/OpenMP crawler, Python document processing, FAISS retrieval, FastAPI RAG, and a Next.js interface.
- Implemented checkpointed HTML/PDF/DOCX cleaning, graph-node and edge reconstruction, document chunking, Sentence Transformer/OpenAI embeddings, and semantic ranking across university resources.
- Generated source-grounded answers with inline citations and graph context, enabling users to trace each response back to original campus webpages and documents.

HealthTrend: Healthcare Big Data Streaming Pipeline

- Built a Dockerized healthcare data pipeline using Apache NiFi, Kafka, Hadoop HDFS, and Spark to ingest structured patient records and stream IoT health events.
- Routed CSV patient data into HDFS and JSON health events into Kafka, then processed both sources with Spark/PySpark for validation and downstream analytics.
- Simulated a healthcare analytics architecture for patient records, heart rate, oxygen level, and temperature data inside an isolated Docker Compose environment.

MarioRL: Super Mario Reinforcement Learning Agent Comparison

- Built a reinforcement learning project comparing PPO and DQN agents on Super Mario Bros using Python, PyTorch, Stable-Baselines3, Gym, and TensorBoard.
- Implemented environment preprocessing with frame skipping, resizing, grayscale conversion, and frame stacking.
- Evaluated agents through checkpoints, training/testing scripts, generated clips, TensorBoard logs, and performance graphs comparing reward, episode length, and training speed.

RAG-Based Analysis of App Update Frequency and User Reviews

- Built a RAG research pipeline using Python, Sentence Transformers, ChromaDB, embeddings, vector search, and review mining to study how mobile app update frequency relates to Google Play review feedback.
- Applied semantic retrieval, sentiment analysis, JSON preprocessing, and review-mining techniques to compare maintenance patterns, recurring user concerns, and app version histories.

RocketGrader: Intelligent Assignment Feedback Platform

- Built a full-stack AI assignment feedback platform with Angular, TypeScript, Express, MongoDB, Auth0, AWS S3, LangChain, and Mistral AI.
- Designed REST APIs and MongoDB/Mongoose models for users, classes, assignments, submissions, files, and chat while supporting role-based teacher/student dashboards.

- Implemented submission upload, parsing, and AI grading across PDF, DOCX, HTML, and text files, returning structured scores, feedback, strengths, improvements, and rubric breakdowns.

Self-Adaptive Parallelism via OpenMP Runtime Coordination

- Built a C++/OpenMP adaptive runtime research system that used per-epoch telemetry and a UCB controller to tune scheduling policy, chunk size, and thread count.
- Benchmarked adaptive scheduling against serial and fixed-policy baselines across Mandelbrot, heat diffusion, and reduction workloads.
- Generated paper-ready performance summaries and plots for runtime, speedup, efficiency, Karp-Flatt behavior, and scheduler decisions.

TECHNICAL SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, C++, C, C#, PHP, Lucee/CFML, ColdFusion

Frontend: React, Next.js, Next.js 15, Next.js App Router, Angular, Vite, PWA, React Native, Expo, Expo Router, React Navigation, Tailwind CSS, DaisyUI, Bootstrap, HTML5, CSS, D3, TopoJSON, react-simple-maps, Framer Motion, Lucide React, React Icons, React Toastify, React Markdown, responsive UI, browser camera APIs, Web Speech API, modal/tab UI patterns

Backend, APIs & Auth: Node.js, Express.js, Flask, REST APIs, Axios, CORS, dotenv, auth middleware, JWT, RBAC, SSO, Auth0, Firebase Auth, Firebase Admin, Drizzle ORM, Mongoose, bcrypt, @react-pdf/renderer, Lucee/CFML backend workflows

Cloud, Data Engineering & DevOps: AWS Glue, AWS Glue Workflows, AWS S3, AWS Lambda, AWS IAM, AWS DynamoDB, AWS SDK for JavaScript, AWS Rekognition client code, PySpark, Spark, Apache NiFi, Apache Kafka, Apache Zookeeper, Hadoop HDFS, MapReduce, Impala, ETL, reverse ETL, data validation, data profiling, MDM, Semarchy MDM/xDM, partitioned S3 pipelines, golden tables, JDBC, Docker, Docker Compose, devcontainers, WSL2, Colima, Git, Linux, Make, Vercel, AWS Amplify

Databases: PostgreSQL, MySQL, MongoDB, MongoDB Atlas, Oracle, ChromaDB, Firebase/Firestore

AI, ML & Retrieval: AI/model gateways, LiteLLM, smart-router tiers, guardrails, Langfuse tracing, Slack block alerts, model routing, model registry design, cost/rate controls, fallback chains, decision records, Codex, Claude Code, OpenAI API, Gemini API, ElevenLabs API, Featherless.AI, Llama 3.1, Qwen2.5-VL, Mistral AI, LangChain, RAG, prompt engineering, LLM inference, multimodal reasoning, automated LLM evaluation, HuggingFace, Sentence Transformers, embeddings, vector search, semantic retrieval, cosine similarity search, PyTorch, TensorFlow, Stable-Baselines3, reinforcement learning, PPO, DQN, Gym, TensorBoard, scikit-learn, NLP, review mining

Security, Testing & Compliance-Aware Engineering: SSO, JWT, RBAC, auth middleware, anti-CSRF tokens, audit logging, IP-aware logging, access-denied workflows, protected 404 workflows, data access controls, PII tokenization, hashlib, boto3, healthcare/compliance-aware workflows, Vitest, Testing Library, React Testing Library, ESLint, route tests, component tests, architecture guardrail tests, Firebase rules testing, project-management API integration

Cybersecurity Education Tooling: NICE Framework, CAE-CD curriculum mapping, CyberGuardian, Kali Linux, OSINT, Nmap/NSE, Metasploit, OpenVAS/Greenbone, Wireshark/tcpdump, Burp Suite, OWASP ZAP, OWASP Top 10, Shodan, theHarvester, Netcat/Ncat, Masscan, Snort, John the Ripper, Hydra, DVWA, Metasploitable2, Vulnserver, PentesterLab, SPF/DKIM/DMARC, PGP/GnuPG, phishing analysis, cryptography, PKI, secure key management